

The SHallow RADar (SHARAD) Onboard the NASA MRO Mission

The Mars Reconnaissance Orbiter, looking for traces of life on Mars, seeks traces of water on or below the Martian surface.

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ABSTRACT | This paper describes the mission concepts, design, and achievements of the Italian Space Agency (ASI)-provided Mars SHallow RADar (SHARAD) sounder high-frequency (HF) sounding radar, used onboard the National Aeronautics and Space Administration (NASA) Mars Reconnaissance Orbiter (MRO) Spacecraft. Its goals are the detection of liquid or solid water below the surface, and the mapping of subsurface geologic structures. Following a brief overview of the MRO mission and of its main science objectives, the paper introduces the basic principles of operation of the radar sounder, and addresses the major design issues faced by such a system. The greatest challenges faced in the design are the control of the interference from off-nadir echoes and the need for a high signal fidelity over a very large fractional bandwidth. The core of the paper is devoted to describing how the above problems have been tackled in the design of the SHARAD instrument, and the main characteristics of its architecture. The two key features of the instrument system design are 1) generation of the transmitted signal directly at the transmitted frequency; and 2) sampling performed directly at the radio frequency (by means of a subsampling technique). The careful design of these features, intended to keep the analog signal path very simple, minimizes distortions

and stability problems. An overview of the calibration approach of both the system impulse response and the antenna gain at nadir versus solar array position, an assessment of the in-flight performance of the instrument, and a short summary of the achieved science results are also provided.

KEYWORDS | Calibration; clutter; ground penetrating radar (GPR); spaceborne radar

I. INTRODUCTION

The Mars Reconnaissance Orbiter (MRO) spacecraft, a National Aeronautics and Space Administration (NASA)/Jet Propulsion Laboratory (JPL) mission, represents a significant step forward in exploration capabilities with respect to the previous Martian orbiters: it is larger, carries more scientific instruments, and can download to Earth a much higher volume of data than any of its predecessors [1].

The search for water, either in liquid or solid form, is a very high priority in the international Mars exploration program and hence for this mission in particular. The formal science objectives identified for the mission are to:

- characterize the present climate of Mars;
- determine the nature of complex layered terrain on Mars and identify water-related landforms;
- search for sites showing evidence of aqueous and/or hydrothermal activity;
- identify and characterize sites with the highest potential for landed science and sample return by future Mars missions.

To achieve these objectives, MRO carries five science instruments, including the Italian Space Agency (ASI) ground penetrating SHallow RADar (SHARAD), devoted

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to the task of detecting the possible existence of liquid or solid water in subsurface (underground) reservoirs.

The design requirement imposed by ASI together with the instrument science team is the detection of interfaces as deep as 1 km between soil and liquid water or water ice.

Transmission to the Earth of the data from the instruments is ensured by a wideband X-band downlink channel, sided by an experimental Ka-band channel. The requirements of onboard data reduction by the instruments have been minimized by the high downlink capability (up to 4 Mb/s) of the spacecraft.

MRO, developed by Lockheed Martin Space Systems under contract to NASA/JPL, was launched on August 12, 2005 from the Kennedy Space Center in Florida by an Atlas V-Centaur launch vehicle, and reached Mars orbit seven months later. An additional six months were required to change the initial, highly elliptical orbit to the operational orbit at around 300-km altitude.

In order to reach the operational orbit, MRO utilized a maneuver called “aerobraking.” This technique is based on the use of the aerodynamic drag during passages in the Mars upper atmosphere to reduce the spacecraft velocity at the periapsis, therefore lowering the altitude at apoapsis. Aerobraking is an innovative maneuver capable of achieving an optimal orbit with minimum usage of fuel. At the same time the design of this maneuver is extremely critical and must be executed very gradually over several months.

The operational orbit was reached in late August 2006. It is a quasi-circular, sun-synchronous, nearly polar orbit that provides overflight of almost the entire planet, with a slight backward inclination to keep the orbit plane fixed with respect to the Sun. The result of this choice is that all passages occur at the same local time, selected for the optimal illumination conditions for the operation of the optical instruments.

At the time of the writing of this paper, MRO was close to completing its fourth year of operation, with all its instruments working normally. The baseline mission duration is four years (two years in the “main science phase” plus two in the “relaying phase”) but extension of the mission is envisaged if the spacecraft remains operationally healthy.

II. SHARAD PRINCIPLES AND SCIENCE OBJECTIVES

Low-frequency [high-frequency (HF) and below] radio waves have the capability to penetrate both soil (especially dry soil, due to its low conductivity) and ice. This capability is exploited by ground penetrating radars (GPRs), used for terrestrial applications such as location of buried pipes, archeological surveys, and forensic searches.

A large part of the transmitted radar energy is reflected back at the surface, but a fraction of it propagates into the

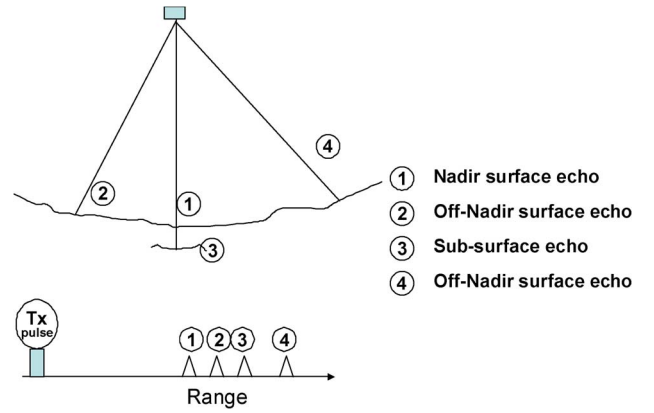


Fig. 1. Scattering from subsurface and off-nadir surface scatterers.

terrain with an attenuation dependent on the wavelength and material dielectric and magnetic characteristics.

Further reflections occur at interfaces between materials with different dielectric constants, thus allowing low-frequency radars to identify different geological layers. Information about the properties of the different layers can be inferred from:

- the velocity of propagation in the medium (which is proportional to $\sqrt{\epsilon}$);
- the attenuation within the medium;
- the fraction of energy scattered at the interface between two media.

The first planetary GPR or “radar sounder” [the Apollo Lunar Sounder Experiment (ALSE)] was placed on the Apollo 17 orbiter with the aim of mapping the Moon’s subsurface [2]. On Mars, where the soil is expected to be very dry (the detection of moist soil would be a remarkable discovery in its own right) much higher penetrations than those achievable on Earth at the same frequencies are expected. This is the principal advantage of such an instrument.

On the other hand, operating from high altitude, with an antenna which necessity is of limited directivity due to the low operating frequency and the allocation constraints on a spacecraft, generates problems of “clutter discrimination” [3]. As can be seen from Fig. 1, together with subsurface echoes from the nadir direction, echoes from surface scatterers in the off-nadir directions are also received. The surface scattering varies in accordance with the surface roughness at the operating wavelength (15 m for SHARAD), and the relevant echoes can be strong enough to be disguised as echoes from subsurface features in the final products.

On a relatively flat surface, the horizontal resolution is related to range resolution (operation in the so-called “pulse limited” mode), according to

$$R_{pl} = \sqrt{\frac{2cH}{B}} \quad (1)$$

where H is the spacecraft altitude; B is the chirp bandwidth; c is the speed of light; and R_{pl} is the radius of the pulse-limited circle.

Over specular surfaces, where coherent reflection can be assumed, the resolution becomes that of the “Fresnel circle,” given by

$$R_F = \sqrt{\frac{\lambda H}{2}} \quad (2)$$

where λ is the transmitted wavelength; and R_F is the radius of the Fresnel circle.

Over complex topographies as found in the Mars environment where SHARAD operates, the above conditions are seldom encountered, leading to a degradation of the horizontal resolution.

The most obvious way to mitigate the problem is to reduce the antenna footprint by increasing its directivity, but this, taking into account the low operating carrier frequency (20 MHz for SHARAD) was not feasible onboard MRO. Increasing the antenna gain (e.g., by using a Yagi or log-periodic antenna) would have required a much larger antenna with a more complex deployment system. In the SHARAD case, this would exceed the volume and mass constraints of the mission, not to mention the increased risk of failure induced by a complex deployment mechanism.

A method, largely employed on both spaceborne and airborne radars to narrow the antenna footprint without the need for a larger antenna, is the synthetic aperture technique. This technique utilizes the relative motion between the radar and the target to improve the resolution in the along-track direction. The principle is relatively simple: an echo exhibits a Doppler frequency shift that is proportional to the transmit frequency and to the radial component of the relative velocity

$$f_D = \frac{2v}{\lambda} \sin \theta \approx \frac{2v}{\lambda} \theta \quad (\text{for small values of } \theta). \quad (3)$$

Targets that are exactly at 90° with respect to the spacecraft velocity vector (i.e., $\theta = 0$) have zero Doppler shift, while closing and receding targets have positive and negative Doppler shift, respectively.

The Doppler shift of a target, seen by the spacecraft, is depicted in Fig. 2, with the Doppler crossing the zero point at which the target is exactly orthogonal to the spacecraft velocity vector.

The Doppler history of each elementary scatterer can therefore be correlated with the ideal one to extract its accurate azimuth position, with a resolution far better than that achievable by the antenna directivity alone, thus “synthesizing” a narrower antenna beam, even if only in the direction of motion.

This significantly reduces the clutter echo energy, but does not eliminate the effect of surface echoes coming

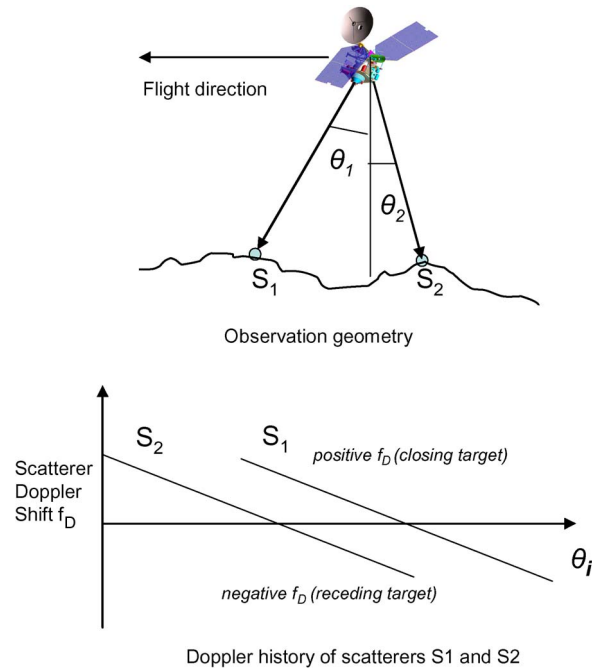


Fig. 2. Target Doppler history.

from the side of the track, which can erroneously be detected as subsurface echoes [4], [5].

The only way to discriminate these clutter returns is to develop a simulator of the radar using a model of the overflown surface (the surface of Mars accurately modeled at the scale of interest), which predicts the presence of clutter echoes. In this way, the corresponding echoes appearing in the measured radargram can be properly interpreted as clutter and then neglected.

To achieve fine resolution in depth, a linear frequency modulated (“chirp”) pulse has been adopted, as in many radar systems. This technique provides the capability of using long pulses [the processed signal-to-noise ratio (SNR) is proportional to the total pulse energy—long pulses then allow one to achieve the same detection capabilities with lower peak power] without jeopardizing the range resolution, which is proportional to the pulse bandwidth. For SHARAD, a 10-MHz bandwidth has been adopted corresponding to a time resolution of 100 ns, equivalent to a range resolution (in free space, two ways) of 15 m [6].

Correlation of the echo with a similar chirp signal (normally referred to as “range compression”) is then performed in the ground processing to achieve the theoretical resolution. The outcome of the range compression is a sinc function with a 6-dB amplitude of the main lobe equal to $1/B$, where B is the chirp bandwidth.

For a given time resolution, the range resolution actually changes with the propagation velocity in the medium, proportional to the square root of its dielectric constant.

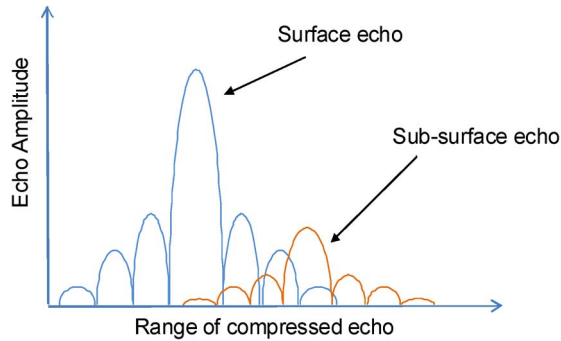


Fig. 3. Strong surface echo masking a weaker subsurface return.

It must be remembered that the $1/B$ resolution is relevant to unweighted pulses. It is customary, in order to reduce the sidelobes of the sinc function, to use amplitude weighting in the compression process. This allows a reduction in the sidelobe amplitude at the expense of a widening of the main lobe and a loss in SNR.

The control of range sidelobes is a very critical aspect of the design of a radar sounder. Fig. 3 illustrates the problem of detecting a weak signal in presence of a strong one. The weak subsurface echo can be hidden by the sidelobes of the surface return, which could be of the same order of magnitude, making it impossible to discriminate layer interfaces close to the surface.

To minimize this effect, heavy weighting is required in the processing. SHARAD ground processing uses a modified Hamming window that is capable of offering strong sidelobe reduction at the expense of a main lobe widening of around 65% (bringing the actual range resolution, in free space, to around 25 m).

Fig. 4 depicts the range sidelobes requirements for SHARAD.

While theoretically the required level of the sidelobes can be easily obtained with a proper weighting function as mentioned above, in the real world, the nonidealities in the transmit and receive chains can jeopardize the theoretical performance. Ripples in the amplitude and phase response can induce paired echoes, which can deteriorate the system impulse response. This is a problem common to all radar systems, and is especially critical for a system like SHARAD due to its large fractional bandwidth of about 50%.

We discuss below how this problem has been tackled in the design of the instrument.

III. MAIN ENGINEERING PARAMETERS

The main engineering parameters of SHARAD [6]–[8] are summarized in Table 1.

The expected ground penetration of the instrument (depending on the nature of the soil) is 1 km, while the

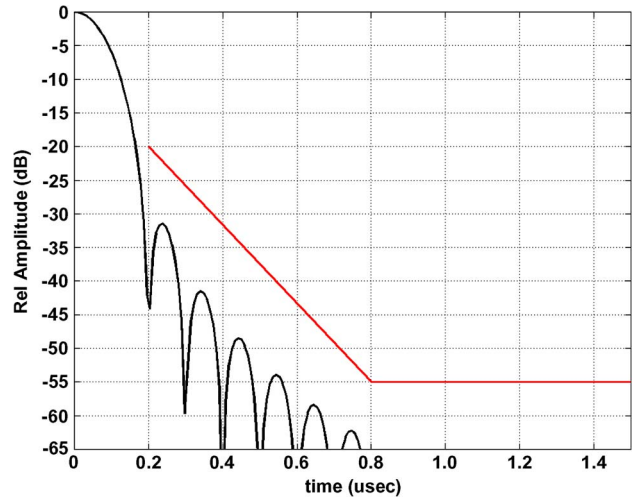


Fig. 4. SHARAD range sidelobe requirements (continuous red line) versus typical response.

along-track resolution, after on-ground synthetic aperture radar (SAR) processing, ranges from 300 to 1000 m.

The instrument nominal pulse repetition frequency (PRF) of 700 Hz has been selected according to the following considerations.

- PRF will be at least twice the maximum expected Doppler shift of the echo with respect to the Nyquist criterion.
- PRF will be low enough to allocate all the expected radar-to-surface range variation due to the orbit altitude range plus topographic margin (this allows use of a fixed PRF, avoiding the additional complexity of adaptively changing the PRF in accordance with the surface range).

Table 1 Instrument Main Parameters

Parameter	Value
Nominal science orbit altitude:	255–320 km
Extended/contingency orbit altitude:	230–407 km
Topographic margin:	–20/+10 km
Centre frequency:	20 MHz
Chirp bandwidth (B):	10 MHz
Pulse width:	85 microsecs
Receive window width:	135 microsecs
PRF (nominal):	700.28 Hz *
PRF (low orbit):	775.19 Hz *
PRF (high orbit):	670.22 Hz *
Radiated power:	10 W
Mass	
SEB (Electronics Box):	11.6 kg
Antenna + cabling:	5.5 kg
Power consumption	
Acquisition:	28W
Std-By:	13W
*) Can work with halved PRF	

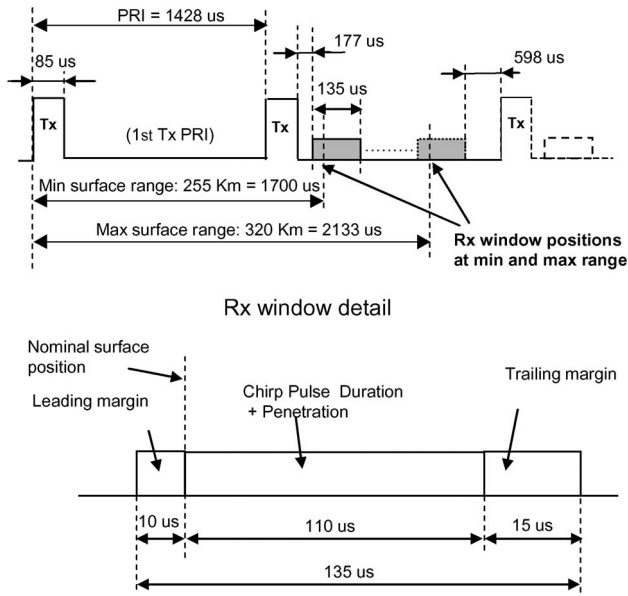


Fig. 5. SHARAD instrument general timing (note: operation with halved PRF is possible in order to operate in rank 0).

In addition, the PRF is selected, together with the pulse width, to maximize the transmit duty cycle and consequently the transmitted energy consistent with the available transmitter peak power.

Fig. 5 depicts the timing for the nominal PRF. With 700 Hz the instrument works in “rank 1,” i.e., the echo from the N th Tx pulse is received after the $(N + 1)$ th pulse. A transmit pulse of 85 μ s (with a resulting duty cycle of around 7%) and a receive window of 135 μ s were introduced in agreement with the PRF selected.

In turn, the width of the receive window is defined in order to allocate:

- the echo from an individual scatterer (85 μ s as the chirp width);
- the subsurface penetration time (25 μ s corresponding to 3.75 km in free space);
- a margin to deal with possible inaccuracies in the surface tracking (10 + 10 μ s of leading and trailing margin);
- an extra trailing margin to cope with delays induced by the ionosphere (5 μ s).

The receive window is dynamically positioned according to *a priori* knowledge of the spacecraft orbit and surface topography (programmed by the mission planning). A closed-loop tracking system is also available.

It is also possible to operate the instrument with a “halved PRF.” This feature has been introduced to avoid possible range ambiguities that can occur from far off-nadir scatterers (around 40°–50° off-nadir angle) in the presence of particularly difficult topographies (at the expense of a 3-dB reduction of the SNR). It has not been needed to date in the operation of the instrument around Mars.

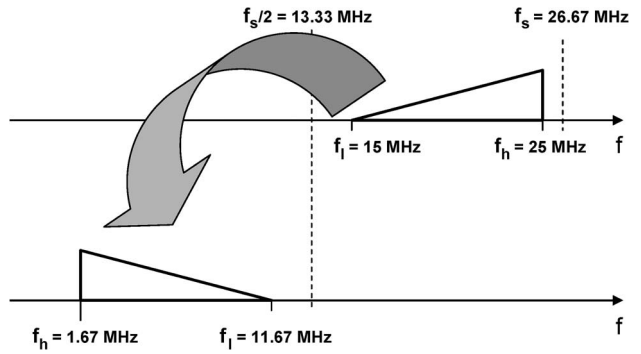


Fig. 6. Spectral folding due to the downsampling process.

Two extra PRFs are also selectable by command to allow operation during an extended mission, in which the spacecraft could be outside the orbit range designed for the primary mission. The halved PRF mode is available for both.

IV. ARCHITECTURE AND OPERATION

A. Overview

The “high fidelity” of the signal reconstruction, in terms of linearity and amplitude/phase distortions (which, in turn, affects the sidelobes of the compressed signal) is obtained mainly by avoiding any upconversion or down-conversion process in the system, and by reducing the analog electronics to a minimum.

The transmit chirp is synthesized directly at the transmit frequency (15–25 MHz) by a digital signal generator and then amplified at the required power level before being sent to the matching network and the antenna.

In the receiver, no analog conversion to baseband is used. The received signal, occupying the frequency range 15–25 MHz, is sampled at 26.67 MHz. The signal is thus folded down around the Nyquist frequency of 13.3 MHz (see Fig. 6), with a resulting digitized spectrum reversed and centered at 6.67 MHz. The oversampling ratio between the minimum theoretical required sampling frequency ($2 \times B$, i.e., 20 MHz) and the actual sampling frequency is 33%. This value is more than acceptable for efficient data transmission on-ground and, therefore, does not need to be reduced by further processing [6]–[9].

Another key feature that contributes to the achievement of very low range sidelobes is the minimum processing performed onboard. Thanks to the large data storage and download capabilities of the MRO spacecraft, SHARAD has no need to perform range compression and synthetic beam formation onboard to reduce the final data rate. The availability of raw or close-to-raw data on the ground gives the user the capability to apply different types of optimized processing to the acquired data, both now and

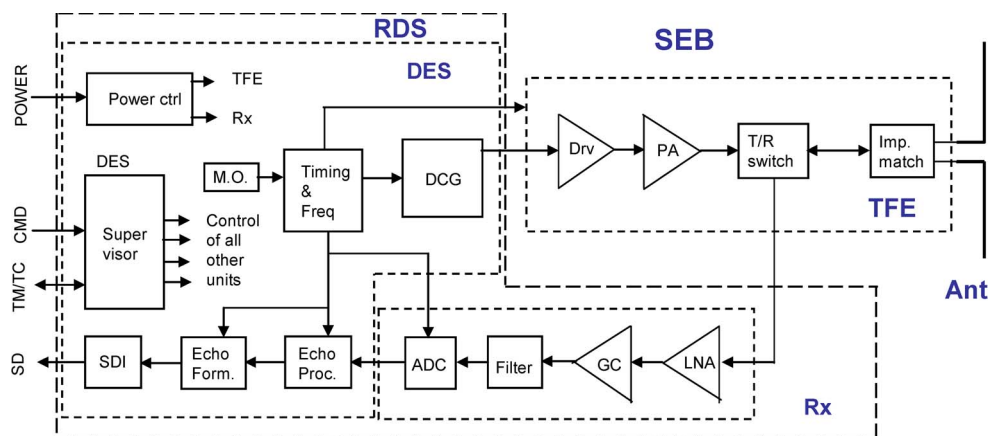


Fig. 7. Block diagram of the SHARAD instrument.

in the future when improved processing techniques may be available. It also allows the range compression to be performed on-ground, using as reference for the correlation not a theoretical chirp but rather an optimized one derived from the calibration performed both on-ground and in-flight to account (and to compensate) for the instrument's nonidealities.

Using this approach, the absolute distortion is not critical: what becomes important, instead, is the variation of the amplitude/phase response of the Tx/Rx chains with respect to the characteristics acquired during calibration, largely performed on-ground.

The minimization of the analog hardware allows a very stable response to be achieved, as a function of both temperature and ageing.

B. SEB Implementation

A general block diagram of the instrument is provided in Fig. 7.

The instrument consists of two main parts: the antenna (a 10-m foldable dipole) and the SHARAD electronic box (SEB). The SEB, in turn, is composed of two main blocks:

- the receiver and digital assembly (RDS), including the controller and DSP functions and the chirp generator [included in the digital electronic subsystem (DES)], and of the Rx module;
- the transmitter and front-end (TFE) in charge of power amplification, antenna matching, and Tx/Rx duplexing.

RDS and TFE are, physically, two separate boxes, mounted inside the mechanical structure of the SEB, which is a table-shaped structure $450 \times 370 \text{ mm}^2$ in area and 190 mm high, whose external side acts as a radiator for the passive thermal control, with the two boxes mounted “upside down” on the inner side (see Fig. 8).

Both RDS and TFE have their internal power converters, and are powered directly by the spacecraft unregulated 28-V power bus.

At the core of the digital chirp generator (DCG in the block diagram) there is a specialized Application Specific Integrated Circuit (ASIC), which is capable of generating a chirp signal fully programmable in central frequency, pulsewidth, and bandwidth.

The digital chirp generator also implements the Doppler compensation function (to compensate for the spacecraft radial velocity): Doppler compensation is actuated by introducing a phase shift between transmitted pulses, exploiting the programmability of the phase accumulator to add a phase offset that is incremented on each pulse. This is equivalent to introducing a frequency shift in the transmit signal to compensate for the Doppler.

The TFE unit is in charge of amplifying the signal from the chirp generator to the high power level required for transmission and for the impedance matching with the antenna.

The power amplification is provided by a dual stage amplifier using bipolar transistors and operated at a regulated 28-V collector voltage, making available an output power of 25-W peak (1-W peak the first stage). The final stage uses a push-pull configuration in class C.

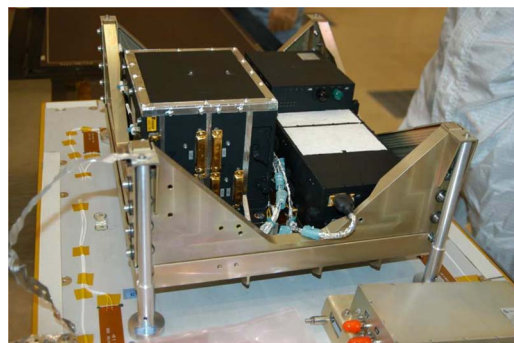


Fig. 8. The SHARAD electronics box (SEB).

The output of the final stage is matched to 50 Ω , and is then followed by a high-power P-I-N diode switch, which acts as a duplexer, and then by the antenna matching network, shared between the Tx and Rx paths.

The matching network maximizes power transfer to/from the antenna, transforming the 50- Ω unbalanced source to the balanced antenna load. The matching must be implemented over a 50% bandwidth with an antenna impedance that is highly variable with frequency. The design selected was a 3-cell low-pass LC network followed by a 4 : 1 transformer for the balanced-unbalanced (balun) conversion.

This provides a return loss better than 6 dB (75% power transfer efficiency) over the whole 15–25-MHz bandwidth.

Additionally, to make the transmitter more robust with respect to mismatched loads (in order to permit the instrument to operate—even if with degraded performance—in case of improper or partial deployment of the antenna) a 2-dB attenuator is added between the final stage and the switch.

The receiver side also uses minimal hardware, providing direct amplification of the received signal followed by analog-to-digital (A/D) conversion directly on the carrier.

Considering that the duplexer switch, located in the TFE, has a limited isolation, the signal that may leak into the receive chain during transmission can drive the receiver itself into saturation. For this reason, at the input of the receiver chain, there is a monolithic single pole, double throw (SPDT) switch, which blanks the receiver during the transmission of the signal. Downstream of the switch, after a first amplifier stage, there is the anti-aliasing filtering section.

The required band shape over the 50% fractional bandwidth of the signal is not obtained with a band-pass filter, but with a cascade of high- and low-pass filters, with another gain stage in between to provide isolation. Both filters are of Chebychev type, the high-pass of the seventh order and the low-pass of the eleventh order.

This section is followed by a thermal compensation network (a P-I-N diode attenuator driven by a temperature sensitive network) designed to minimize Rx chain gain variations over temperature.

The gain control function is implemented by two identical gain/attenuation stages, each with a 4-b digital attenuator (0–15 dB in 1-dB step), providing an overall control of the Rx gain over a 30-dB range (the gain setting is programmed under control of the processor). The other two amplifier stages drive the A/D converter, which performs the conversion (with downsampling) at a frequency of 26.66 MHz.

It must be noted that, in the case of the SHARAD receiver, the noise figure is not the key element of the design since the dominant noise contribution to the SNR comes from galactic background sources (estimated about 20 dB above kT).

After the A/D conversion, the only processing applied to the received signal is a coherent presumming, i.e., an averaging of the signal received over different pulse repetition intervals (PRIs). In this way, the amount of data to be transmitted is reduced by a factor equivalent to the number of presumed PRIs. This operation is performed in hardware by a dedicated field-programmable gate array (FPGA), with variable presumming factors: 1 (no presumming), 2, 4, 8, 16, 28, and 32, to reduce the output data rate. The presumming factor is programmable (together with the number of bits to be transmitted) according to the operating scenario to achieve the minimum data rate compatible with the required data quality. In fact, the presumming averages the phase information over several PRIs with a resulting loss of information that causes an increase of the synthetic antenna sidelobes proportional to the amount of the presumming. The associated impact on data quality depends on the topography of the overflowed region.

Phase coherence during presumming is maintained in the presence of radial (i.e., vertical) velocity components thanks to the Doppler compensation in the chirp generator. Full compensation is achieved only at center frequency, but the residual errors at band edges are such as to not introduce significant losses.

The presumed samples are then packetized with 4-, 6-, or 8-b resolution, using either a fixed or adaptive scaling (to better exploit the available dynamic range), by another FPGA, under the control of the instrument processor, and transmitted to the spacecraft data recorder.

The same processor is also in charge of the instrument commanding and monitoring.

All instrument timings are derived from the same 80-MHz master oscillator, using an FPGA-based state machine.

The position of the receive window can be controlled either by *a priori* knowledge of the SHARAD altitude and surface topography (baseline operating mode), or by a closed-loop tracker (included as backup).

In the former case, the instrument computes the position of the range window from:

- the orbital data (radius, radial velocity, and latitude versus time) provided, before each observation, in a file (generated from the mission Flight Engineering Team) called orbit data table (ODT);
- the topographic profiles of the surface (expressed as a sequence of polynomials, each fitting a portion of the target surface corresponding to an acquisition time of 30 s), provided as parameters loaded into the SHARAD parameter table (PT) before each observation, as a function of the latitude.

The need to separate data handling for orbit and surface data originates from the different levels of planning at which they are produced: SHARAD planning is requested at least two weeks ahead while flight engineering planning updates the orbit data no more

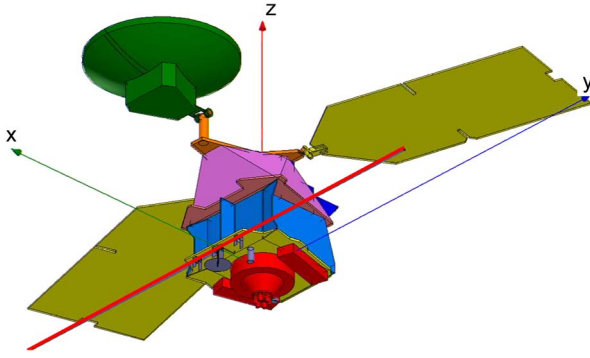


Fig. 9. Antenna installation on the spacecraft.

than one week ahead of the passage to provide the required prediction accuracy.

In the closed-loop mode, SHARAD is able to track the surface using either a center-of-gravity estimator or a threshold detector on the leading edge. In both cases, tracking initialization is performed using the open-loop data mentioned above.

The ODT contains also radial velocity information used to define the Doppler compensation.

C. Antenna

The antenna, manufactured by ASTRO Aerospace (Carpinteria, CA), is a dipole consisting of two fiberglass tubes, each arm 5 m long, acting as a mechanical support structure for a metal wire that runs inside them and that represent the actual “active” part of the antenna.

The antenna tubes, based on the fiber foldable tube (FFT) technology, were folded into five segments each, without using hinges/springs or other mechanical parts for the deployment, but relying only on the material elasticity to extend into the deployed position once released.

During the cruise phase, the antenna was kept stowed by two hinges, and enclosed in a Ge-kapton shield to protect it from the thermal stresses induced by the aerobraking.

The folded antenna had an envelope of $1524 \times 241 \times 190 \text{ mm}^3$.

The antenna installation on the spacecraft is shown in Fig. 9. The dipole is parallel to the y-axis (corresponding to the spacecraft velocity vector), while the Mars surface is in the x-direction.

D. Ground Processing

The ground processing has the primary function of compressing the transmitted chirp waveform and to implement the synthetic aperture to improve the along-track ground resolution.

The range compression is performed by means of an algorithm, called phase-gain algorithm (PGA) [10], that adaptively compensates the phase distortions introduced

by the propagation through the ionosphere and those due to the hardware (mainly the matching network). This algorithm is very well known from the literature and considered reliable in operation conditions similar to those expected for SHARAD. Concerning the synthetic aperture processing, given the unconventional nadir looking geometry, three processing algorithms have been analyzed: omega-k, specan, and chirp scaling or CSA [4].

The choice for SHARAD was based on the following considerations:

- the range cell migration is really significant and requires an accurate correction, for which the CSA algorithm shows the best performance;
- the CSA has focused correctly on the targets in the simulations;
- the CSA has preserved and correctly compensated the Doppler and then the phase history;
- the CSA has a greater computational burden but it is conceptually easy to implement.

For all of these reasons, the final decision has been to implement a CSA adapted to the original nadir looking geometry of SHARAD.

V. CALIBRATION

Calibration is a fundamental step in any science instrument in order to provide valid data. In the SHARAD case, calibration was a must in order also to ensure optimal performance. Two specific aspects need dedicated calibration:

- end-to-end impulse response of the system (to derive the reference chirp for range compression);
- in-flight antenna gain variation with respect to the positions of the spacecraft appendages.

The calibration is divided into two parts: calibration of the system impulse response of the electronics in the SEB (including the matching network), performed on-ground, and in-flight correction of the relevant reference to account for the antenna response [11].

A. System Impulse Response Calibration

For the ground part of the characterization, transmit chirps have been acquired both at lab ambient temperature ($22^\circ\text{C} \pm 2^\circ\text{C}$) and at 20°C steps over the operating temperature range (in thermal-vacuum conditions, representative of the real operating environment) and, in the same way, a “theoretical” chirp has been injected into the receive chain to characterize its end-to-end response. The two have been then combined analytically to generate the “reference” chirp waveform.

During these tests, the antenna was replaced by a balanced dummy load simulating its impedance over frequency, and providing a $50\text{-}\Omega$ test interface with calibrated amplitude and phase response versus frequency.

The reference waveform defined in this way does not take into account the response of the antenna. Therefore, a

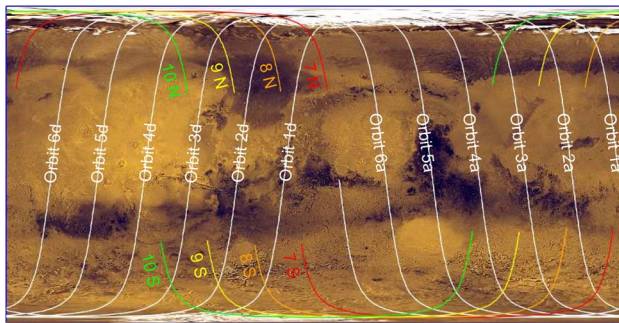


Fig. 10. Orbits during the “calibration day” and relevant crossovers.

correction function has to be retrieved from analysis of echoes from flat surfaces during the commissioning phase, and used to correct the reference waveform.

B. Radiometric Calibration

Significant challenges were also posed by the radiometric calibration of the system, due to difficulties in the characterization of the antenna pattern. The in-flight behavior is indeed strongly dependent on the position of the spacecraft appendages [solar arrays and high-gain antenna (HGA) dish—both moving continuously along the orbit], while the long wavelength (15 m) does not allow to characterize the real hardware using a standard antenna test range. It is worth noting that SHARAD is required to provide calibrated data also with the spacecraft rolled up to $\pm 25^\circ$ (roll is required to allow the HiRiSE instrument to image off-nadir targets), therefore complete radiometric calibration requires knowledge of antenna gain over this axis for $\pm 25^\circ$.

An additional challenge was that only 24 hours were allocated to SHARAD to perform its in-flight calibration.

Preliminary information was collected during the development phase by measurements on a scale model of the spacecraft and the antenna. The scaling factor was approximately 1 : 17 in order to perform the measurements in the frequency range 255–455 MHz. The tests, performed using the outdoor test range of the Rome facility of Thales Alenia Space, were carried out with different configurations of the appendages.

These tests provided preliminary data to be refined later by in-flight calibration, and were also the only reference for absolute antenna gain (only relative calibration was possible in-flight due to the lack of calibrated reference targets on Mars).

To achieve the required accuracy, the in-flight radiometric calibration was then split into the following.

- Calibration of gain in the nominal nadir direction (i.e., spacecraft with zero roll angle) versus appendages configuration (actually, calibration of

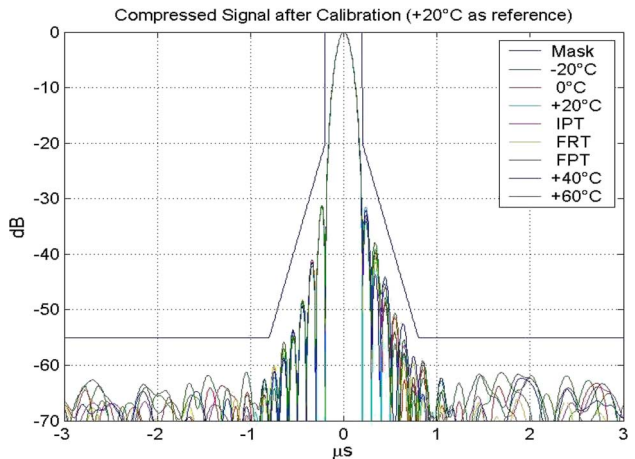


Fig. 11. SHARAD range sidelobes after calibration versus temperature.

the delta with respect to a “reference” configuration). For the purpose of calibration, the virtually infinite configurations of the appendages (six degrees of freedom for the solar array plus two for the HGA) were grouped in four “families” of configurations, with relatively small (< 1 -dB peak) gain variation within a family (using the information from the mock-up tests).

- Calibration of gain versus spacecraft roll angle (performed at the “reference” appendages configuration).

The latter calibration was needed because SHARAD was required to perform acquisitions also while MRO is rolled to image off-nadir targets with the optical instruments. Additionally, the peak of the antenna gain was not in the nadir direction but about 25° away from it.

The first type of calibration exploited the crossovers between ascending and descending orbital tracks (see Fig. 10). In this way, the same target area was overflown twice during the calibration day, once in the reference configuration and once in the configuration to be calibrated. Using the first six orbits of a day for the reference acquisition and another four for the configurations to be calibrated, a total of 32 crossover points (13 for each polar region, and six at the equator) were available in a day. This allowed the minimum objective of three pairs of acquisitions for each configuration (the minimum needed for a reasonable averaging and evaluation of the data dispersion) to be easily achieved.

The latter calibration used the orbit periodicity: the same tracks were repeated every 17 days, with roughly several kilometers of slip. Both passages were performed with the spacecraft in “reference” appendage configuration, but the first in nominal attitude and the second with the spacecraft rolling by $\pm 25^\circ$ along the track at a constant roll rate. The difference between the two observations provided the nadir gain versus roll function.

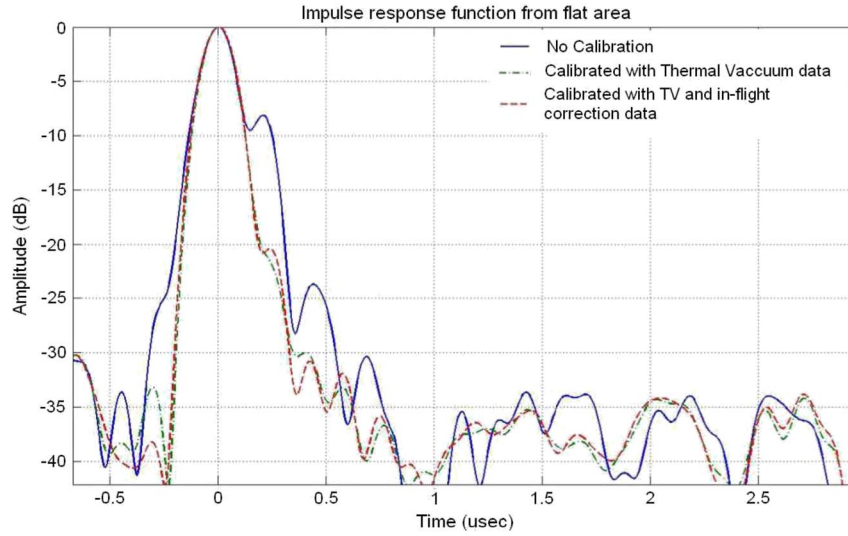


Fig. 12. SHARAD calibrated and uncalibrated surface returns.

The pattern calibration confirmed that, due to the interference from the spacecraft solar array, the peak of the antenna pattern is far off the nominal nadir direction, showing a peak of the response at a roll angle of $+25^\circ$, with a two-way gain of $+3$ dB with respect to the reference zero roll, and -1 dB for a roll angle of -25° .

It should be noted that this actually means that an improvement of the SNR of 3 dB can be achieved if the observation is performed with the spacecraft rolled by $+25^\circ$. Even if this is not a standard procedure due to the resulting significant interferences with other spacecraft operations, it is actually done for specific targets of high interest when the science demands that the instrument be configured for its best possible performance.

Concerning the calibration for the spacecraft configuration, variations from -3.4 to $+3.8$ dB with respect to the “reference” configuration have been measured. These values are in line with those predicted by ground tests on the scaled model.

All the calibrated parameters (reference function versus temperature, and absolute gain correction with respect to temperature, attitude, and configuration) were archived in a calibration database and used in the data processing to deliver fully calibrated data products.

With this approach, a relative calibration accuracy better than ± 2 dB has been achieved.

VI. PERFORMANCE

The system detection capability performance is driven by its range sidelobe level. With our approach of an on-ground correlation using a calibrated reference function, the stability of the Tx/Rx chains plays a major role in achieving this goal.

While stability over the mission lifetime cannot be practically measured before the mission, the stability with temperature (which is accurately characterized before launch and compensated in the ground segment) can be measured with a sufficient degree of accuracy and can provide insight on the system robustness.

Fig. 11 shows the system impulse response at different temperatures, but always using the reference function collected at lab ambient temperature. The excellent behavior (well below the required mask) over temperature provided an indirect indication of the weak sensitivity of the frequency response to component drifts. Hence we assume that the on-ground characterization data will be valid for a good in-flight data correlation through the

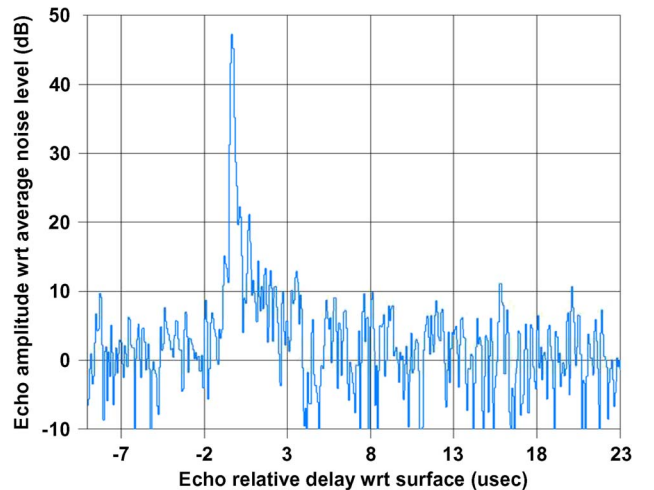


Fig. 13. SHARAD range line acquired over flat area.

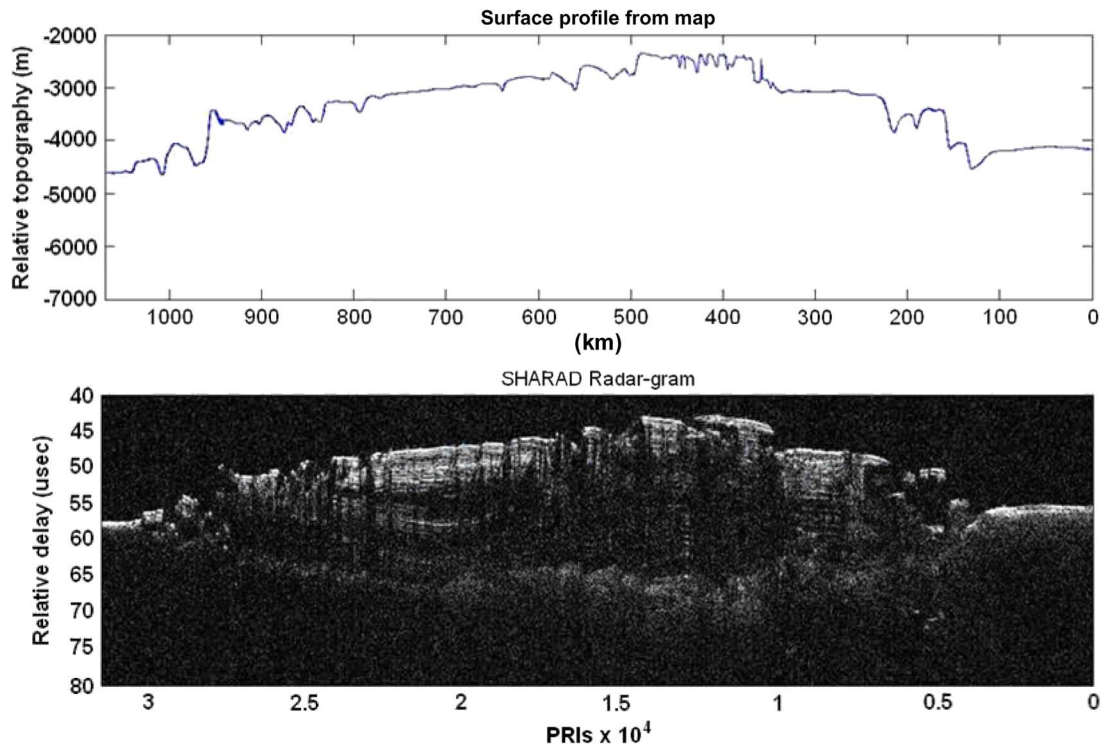


Fig. 14. SHARAD radargram (bottom) compared with the surface profile derived from the available Mars altimetric map (top). The radargram shows the echo intensity (represented as the brightness of each point) as a function of echo delay (y-axis) and along-track distance (proportional to the number of PRIs, x-axis). The first echo from the top, in each pulse repetition interval, following the surface profile, is the echo from the surface. The echoes at larger delays are echoes at the interface between subsurface layers.

whole operating life of the instrument. This assumption was confirmed during the initial tests of the instrument performed in Mars orbit, more than two years after the characterization data were collected [12]. Fig. 12 shows an echo from flat surface with three different types of processing:

- without calibration applied;
- using only calibration data collected on-ground;
- using a reference function incorporating antenna calibration.

Thanks to the system impulse response calibration, the widening of the main lobe and the increase of the level of the first sidelobes, induced by amplitude and phase distortions, were kept effectively under control, thus preserving the ability of the instrument to resolve scatterers close to the surface.

Fig. 13 shows another range cut, acquired over a flat region with no significant subsurface returns, as an example of the dynamic range achievable. A few microseconds after the surface echo, the dynamic range is limited by the system noise, caused by galactic noise plus the electromagnetic interference from spacecraft equipment, which also gives a significant contribution to the total noise.

The resulting dynamic range is of the order of 47 dB. It should be noted that, during this acquisition, the spacecraft was not rolled for the optimum antenna gain. Such a roll would have allowed for an extra 3 dB of SNR.

An example of acquisition is provided in Fig. 14: the SHARAD processed radargram is in the bottom figure, while the altimetric profile of the overflow region (coming from the Mars altimetric maps) is depicted in the top.

The radargram was acquired on the northern polar region of Mars and shows, in a very clear way, the sequence of layers (interpreted to be alternating layers of water ice and dust) close to the surface. Layers as close as few tens of meters can be clearly resolved. The azimuth resolution of the radargram, obtained by the on-ground azimuth processing, is 300 m.

VII. SCIENCE RESULTS SUMMARY

The SHARAD scientific results are described in several papers published on different international journals [13]–[24].

A few of the most important are summarized here to give the reader a flavor of achievements produced so far by the SHARAD instrument.

Martian surface features identified as lobate debris aprons (LDAs) are thick (hundreds of meters) masses of material that extend up to several tens of kilometers from high relief slopes and terminate in lobate fronts. Their geomorphic expression and restricted occurrence in latitude has led numerous workers to conclude that LDAs contain water ice, but the suggested amount of ice involved in their formation and evolution has ranged from minor interstitial ice in rocky talus to predominantly ice in debris-covered glaciers. SHARAD data have provided the evidence that LDAs in the Deuteronilus Mensae region of the mid-northern latitudes in fact do consist mostly of ice.

The south polar layered deposits (SPLD) of Mars have been studied through imagery for decades. Now the subsurface sounding performed by the ASI/NASA radar MARSIS on the European Space Agency Mars Express Orbiter and SHARAD have provided data at multiple frequencies (1.8–5 and 20 MHz, respectively). Both instruments detect subsurface reflections in the Promethei Lingula region of the SPLD. In particular, SHARAD detects tens of reflections without penetrating to the base of the SPLD, which is detected by MARSIS. The joint analysis of the radar data sets confirms several predictions concerning the interior of the SPLD from stratigraphic studies of images, including that most of the layers extend throughout the region and that they decrease in elevation toward the margin of the SPLD.

Sounding radar profiles across Mars' Amazonis Planitia reveal a subsurface dielectric interface that increases in depth toward the north along most orbital tracks. The maximum depth of this detection is 100–170 m, depending upon the real dielectric permittivity of the materials, but the interface may persist at greater depth to the north if the reflected energy is attenuated below the SHARAD noise floor. The dielectric horizon likely marks the boundary between sedimentary material of the Vastitas Borealis Formation and underlying Hesperian volcanic plains, representing two distinct epochs of ancient Martian history.

The SHARAD-detected interface follows the surface topography across at least one of the large wrinkle ridges in north central Amazonis Planitia. This feature suggests that Vastitas Borealis sediments, at least in this region, were emplaced prior to a period of strong compressional tectonic deformation. The change in radar echo strength with time delay is consistent with a loss tangent of 0.005–0.012 for the column of material between the surface and the reflector. These values are consistent with dry, moderate-density sediments or the lower end of the range of values measured for basalts. Other observations taken over the Gemina Lingula region, one-fourth of the area of the north polar layered deposits, show a drop of the dielectric constant that could be explained by an abrupt 250-m uplift of the base. The bulk ice in the region studied has an average dielectric constant of 3.10 and a loss tangent < 0.0026 , consistent with the hypothesis that the volume of the observed ice is pure to the 95% level.

VIII. PRESENT OUTLOOK AND LESSONS LEARNED

As of this writing a mission extension has been granted for MRO. SHARAD will therefore continue to collect science data for some years to come. Similarly ESA's Mars Express mission has been extended and thus MARSIS will continue to operate around Mars.

The design choices implemented in SHARAD provided to the science team a very powerful tool of great importance for the understanding of the Mars geology and stratigraphy. Along with its companion instrument MARSIS, SHARAD has paved the way for a wider usage of radar sounders in planetary exploration. Possible missions with radar sounders are currently being discussed for the exploration of the icy moons of Jupiter and Saturn, and also for probing the internal structure of comets.

Today, on the basis of the experience gained with SHARAD, some development trends for future sounders are under study.

- Multifrequency operation, to complement the high penetration possible at lower frequencies, with the better resolution possible with the higher frequencies.
- Improved antenna systems—exploiting the technology in deployable/foldable structures to manufacture more complex, larger deployable antennas capable of providing better directivity and gain while remaining in the same envelopes of mass and stowage volume.
- Onboard processing, such as range and azimuth compression (already used on MARSIS but not on SHARAD) will be required to reduce the generated data rate for the more distant planets Jupiter and Saturn. Data rate limitations afforded by the greater distances have been a significant factor in planning outer solar system missions.
- Simple, almost all-digital radio-frequency (RF) architectures will guarantee better data fidelity. More flexible, adaptive processing and data compression will be used to maintain high data quality while minimizing the volume of data to be downloaded. ■

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A. Safaenili, who followed the SHARAD Instrument at NASA/JPL since its early phases, has contributed enormously to this paper and to the project. His support during both the design and the operation phases was invaluable. A sudden and terrible disease took him away from his work, his family, and his friends in July 2009, while the draft of this paper was being sketched out. We mourn his loss and wish he were still here to be an author and collaborator. We dedicate this paper to his memory.

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Roberto Seu was born on February 18, 1959. He received the M.S. and Ph.D. degrees from the "Università degli Studi La Sapienza," Rome, Italy in 1985 and 1990, respectively.

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